

ANSWER KEY



- 10 Find the inverse of $f(x) = 4x - 8$.
Swap x and y in $y = 4x - 8$: $x = 4y - 8$, so $y = \frac{x+8}{4}$. $f^{-1}(x) = \frac{x+8}{4}$.
- 11 Find the inverse of $f(x) = \frac{x-3}{2}$.
Swap: $x = \frac{y-3}{2}$, so $2x = y - 3$, giving $y = 2x + 3$. $f^{-1}(x) = 2x + 3$.
- 12 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.
 $f(g(x)) = 5\left(\frac{x-1}{5}\right) + 1 = (x-1) + 1 = x$, confirming they are inverses.
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Transformations of graphs

- 13 The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f .
 $y = f(x+4) - 2$.
- 14 If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g .
 g reflects f over the x -axis, then shifts the result up 3 units.
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Piecewise functions

- 15 A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.
 $f(-3) = -3 + 2 = -1$ (using the first piece, since $-3 < 0$). $f(3) = 3^2 = 9$ (using the second piece, since $3 \geq 0$).
- 16 A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.
 $f(2) = 2(2) = 4$ (using the first piece, since $2 \leq 2$). $f(5) = 5 + 5 = 10$. As $x \rightarrow 2$ from the right, $f(x) \rightarrow 2 + 5 = 7$, which does not match $f(2) = 4$, so f is not continuous at $x = 2$ — there is a jump.
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Reading function graphs

- 17 The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x -intercepts and y -intercept, then find them exactly.
 x -intercepts: solve $x^2 - 4 = 0$, giving $x = 2$ and $x = -2$. y -intercept: $f(0) = -4$.
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Even, odd, and other function properties

- 18 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.
 $f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so f is even.

- 19 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

$f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.

A well-designed word problem: modeling a car rental cost

- 20 This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

(a) $C(5) = 45(5) + 20 = 245$. Renting the car for 5 days costs 245 dollars total. (b)

$P(d) = 45d + 20 - 200 = 45d - 180$. $P(5) = 45(5) - 180 = 225 - 180 = 45$ dollars. (c) From $x = 45d + 20$, solve for d : $d = \frac{x-20}{45}$, so $C^{-1}(x) = \frac{x-20}{45}$. For $x = 335$: $d = \frac{335-20}{45} = \frac{315}{45} = 7$ days.